

Windows Build Status

What is proven to work on native Windows right now, what still needs a real machine to confirm, and the exact steps to build and run it.

01 Status

BUILD

Verified

`forward_search/`'s C++/OpenCL code compiles and links on native Windows via MinGW-w64 (MSYS2), confirmed by GitHub Actions.

RUN

Not yet verified

GitHub's Windows runners do not reliably have a usable OpenCL device, so CI only proves the build compiles and links, not that it runs correctly against real data.

- ① The convenient single command workflow (`projection-center pipeline --backend cpp, run_all_128.sh`, and so on) has not been tested on Windows. Only `forward_search/`'s own C++ build was attempted this session. `projection-center` is a separate Python package (with a `pyopencl` dependency) that has not been touched for Windows at all.

02 Building

Using MSYS2's MinGW64 environment, the same toolchain the CI workflow uses.

```
# from an MSYS2 MINGW64 shell
pacman -S --needed mingw-w64-x86_64-toolchain mingw-w64-x86_64-meson \
  mingw-w64-x86_64-ninja mingw-w64-x86_64-hdf5 mingw-w64-x86_64-boost \
  mingw-w64-x86_64-opengl-headers mingw-w64-x86_64-opengl-icd

# MSYS2's opengl-headers package does not include the C++ bindings header
# forward_search/ uses, fetch it once, matching the CI workflow:
curl -sSL -o /mingw64/include/CL/opengl.hpp \
  https://raw.githubusercontent.com/KhronosGroup/OpenCL-CLHPP/main/include/CL/opengl.hpp

cd forward_search
meson setup builddir
meson compile -C builddir
```

Produces `builddir/forward_search.exe` and `builddir/forward_search_resample.exe`, same as `forward_search` and `forward_search_resample` on Linux.

03 Running (search only)

Same flags as Linux, per `RUNNING.md`.

```
./builddir/forward_search.exe --data path\to\data.hdf5 --mode buffer \
  --kernel kernels/forward_search.cl --output pose.h5
```

Prints the same MSE: / xshift: / alpha: / beta: summary as the Linux binary, assuming a real OpenCL device is present.

04 Full pipeline (search + resample, one command)

Needs `projection-center` installed (`pip install -e projection-center`).

```
projection-center pipeline --backend cpp --cpp-mode buffer --data data.hdf5 \  
  --output-pose pose.json --output-data resampled.hdf5
```

Not yet confirmed working on Windows. CI now attempts exactly this command against a small synthetic dataset, that is the next thing to check.

05 What is still open

- `run_all_128.sh` / `run_all_512.sh` call `python3`, which a plain Windows Python install usually does not provide, just `python`. Would need fixing, and need Git Bash or MSYS2 to run at all.
- Built with MinGW-w64, not MSVC. Switching to MSVC would need further flag changes (different optimization and linker syntax) not attempted yet.

06 Command reference (Windows)

Not identical to Linux, run these instead of copying the Linux commands.

Setup, once, from an MSYS2 MINGW64 shell

```
pacman -S --needed mingw-w64-x86_64-toolchain mingw-w64-x86_64-meson \
  mingw-w64-x86_64-ninja mingw-w64-x86_64-hdf5 mingw-w64-x86_64-boost \
  mingw-w64-x86_64-opengl-headers mingw-w64-x86_64-opengl-icd
curl -sSL -o /mingw64/include/CL/opengl.hpp \
  https://raw.githubusercontent.com/KhronosGroup/OpenCL-CLHPP/main/include/CL/opengl.hpp
cd forward_search
meson setup builddir
meson compile -C builddir
```

Install projection-center (normal Windows Python, not MSYS2's)

```
pip install -e projection-center
```

--backend cpp needs MinGW's runtime DLLs on PATH

```
set PATH=C:\msys64\mingw64\bin;%PATH%
```

List devices

```
projection-center devices
```

Search only (--backend: opengl, cpu, or cpp)

```
projection-center search --backend opengl --data data\projs_change.hdf5 --output-pose pos
```

Resample only (needs a pose.json from a search step above)

```
projection-center resample --backend opengl --data data\projs_change.hdf5 --pose pose.js
```

Full pipeline, search and resample in one command

```
projection-center pipeline --backend cpp --cpp-mode buffer --data data\projs_change.hdf5  
--output-pose pose.json --output-data resampled.hdf5
```

Raw forward_search.exe, search only

Same as --backend cpp above but without installing projection-center. Cannot be chained into resample by hand, same limitation as Linux.

```
forward_search\builddir\forward_search.exe --data data\projs_change.hdf5 --mode buffer \  
--kernel forward_search\kernels\forward_search.cl --output pose.h5
```

❗ No Windows port of run_all_128.sh / run_all_512.sh yet, run the commands above manually in sequence instead.